

## Functional Requirements

| Req ID | Type       | Description                                                                                                                                                  | Priority | Acceptance Criteria                                                                                                                         | Rationale (short)                                                       |
|--------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| FR-001 | Functional | The system shall allow students to register and log in using their credentials, and authenticate them on each session.                                       | High     | Pass: valid credentials grant access to profile dashboard. Fail: invalid credentials rejected with error message.                           | Secures access and personalizes each student's data.                    |
| FR-002 | Functional | The system shall display internship listings posted by admin and allow students to apply to a listing of their choice.                                       | High     | Pass: student's application is recorded and visible under "My Applications". Fail: duplicate applications to the same listing are blocked.  | Core feature enabling students to act on internship opportunities.      |
| FR-003 | Functional | The system shall allow students to practice aptitude, technical MCQs, and coding problems from a categorized question bank.                                  | High     | Pass: student selects a category and receives relevant questions with a scored result. Fail: no questions returned or scoring is skipped.   | Directly supports the portal's core preparation purpose.                |
| FR-004 | Functional | The system shall allow students to take timed mock tests and view their results and detailed answer analysis afterward.                                      | Medium   | Pass: test auto-submits at time limit and shows score + correct/incorrect breakdown. Fail: test allows unlimited time or shows no analysis. | Simulates real interview/test conditions and gives actionable feedback. |
| FR-005 | Functional | The system shall generate a personal analytics summary (performance trends, strengths/weaknesses) for each student based on their test and practice history. | Medium   | Pass: analytics dashboard reflects updated scores after each new attempt. Fail: dashboard shows stale or incorrect aggregated data.         | Helps students track progress and focus preparation effectively.        |

## Non-Functional Requirements

| Req ID  | Type           | Description                                                                                                                                      | Priority | Acceptance Criteria                                                                                                                                     | Rationale (short)                                          |
|---------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| NFR-001 | Non-Functional | The system shall respond to user actions (page loads, test submissions, search queries) within 2 seconds under normal load.                      | High     | Pass: 95% of requests return within 2s in testing. Fail: noticeable lag or timeout on core actions.                                                     | Ensures a smooth, non-frustrating user experience.         |
| NFR-002 | Non-Functional | The system shall protect user data (credentials, personal info, scores) using encrypted storage and secure authentication.                       | High     | Pass: passwords stored hashed; data transmitted over HTTPS. Fail: plaintext storage or unencrypted transmission found.                                  | Protects student privacy and builds trust in the platform. |
| NFR-003 | Non-Functional | The system shall be accessible and function correctly across common browsers (Chrome, Firefox, Edge) and device sizes (desktop, tablet, mobile). | Medium   | Pass: core features usable without layout breakage on tested browsers/devices. Fail: broken UI or non-functional features on any listed browser/device. | Maximizes reach since students use varied devices.         |